ANSYS RedHawk™ Power Integrity and Reliability Analysis

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1 Purpose of This Document

The goal of this document is to enable a user to achieve functional safety compliance satisfying ISO26262 of TCL1 (Tool Confidence Level 1) in Power Integrity and Reliability, using software product ANSYS RedHawk™. This document is a report classifying used tools according to their confidence need during the use of a software tool chain or parts of a software tool chain. The document consists of use-cases, and error scenarios which may happen during the analysis and measures for detecting the tool errors/malfunctions through FMEA (Failure Models and Effects Analysis). We recommend that the user read respective manuals of the product for additional information beyond this safety manual.

- RedHawk™: RedHawk_UserMan_19.2.pdf

2 Terms and Definitions

All abbreviations used in the document and full name of each abbreviation.

Abbreviated Terms	Full Name
CPM	Chip Power Model
DEF	Design Exchange Format
DVD	Dynamic Voltage Drop
EM	Electromigration
FSDB	Fast Signal Database
GDSII	Graphic Data System II
GSR	Global System Requirements
IR	Static Voltage Drop
LEF	Library Exchange Format
LIB	Liberty files
PDN	Power Delivery Network
PI	Power Integrity
PLOC	Pad Location Information
SPEF	Standard Parasitic Exchange Format
STA	Static Timing Analysis
VCD	Value Change Dump
VRM	Voltage Regulator Module

3 ANSYS RedHawk™- Power Noise and Reliability Analysis

3.1 Introduction

ANSYS's RedHawk™ is a full-chip cell-based power/ground design and verification product, addressing power integrity with static and dynamic analyses, power net electromigration analysis, signal net electro-migration and electrostatic discharge, from early in the design flow through verification and sign-off as shown in the Figure 1.

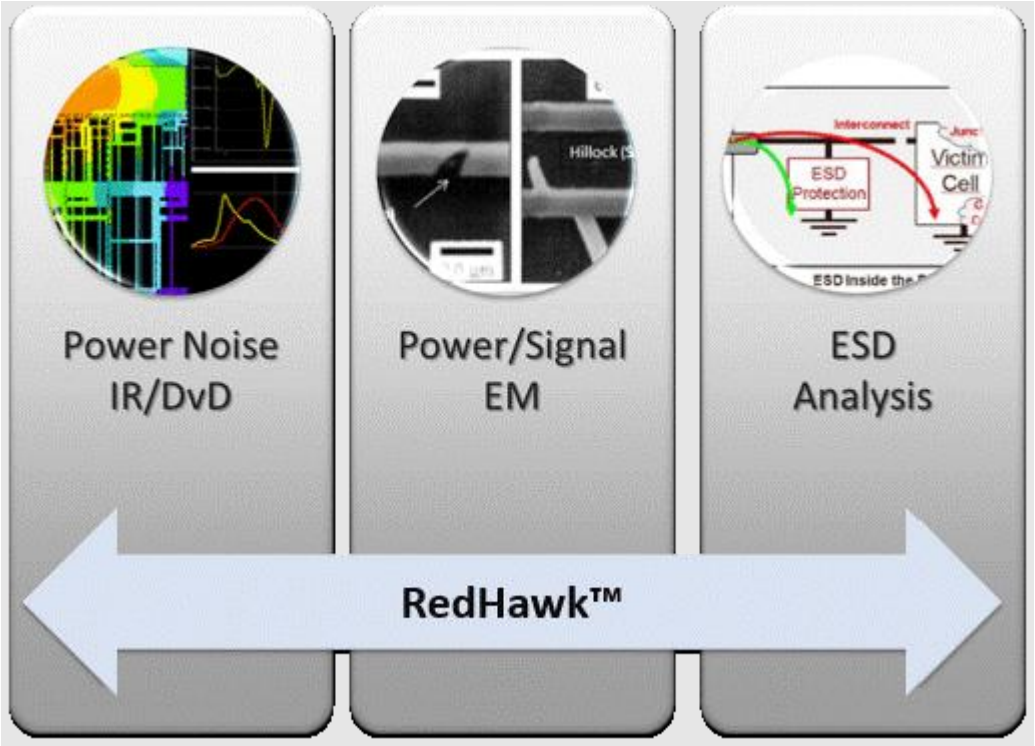


Figure 1: RedHawk™ IR drop, Electromigration and Electrostatic Discharge (ESD) Flows

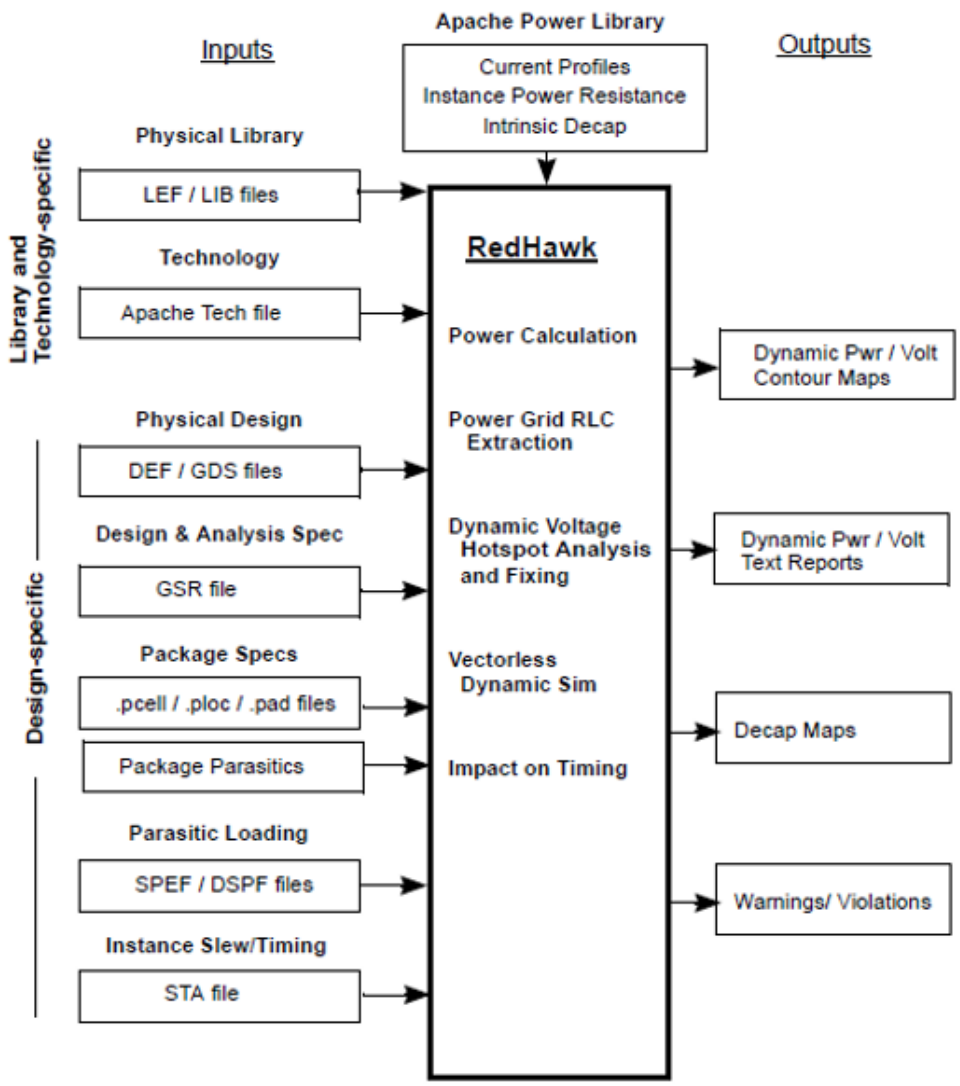
3.2 Product Version

- RedHawk 2019R2

3.3 RedHawk- Design Input Data Requirements

RedHawk™ uses industry-standard design input data for analyzing power noise and reliability on all types of system-on-chip (SOC) digital designs. The design data needed are described in the following table.

File	Description
DEF (*.def*)	Physical description of instances, power and ground, signal network, and other circuit elements
LEF (*.lef)*	Library Exchange Format file. Physical description of library cells
LIB file (*.lib*)	Synopsys-format library files containing logical description of library cells, with power and timing tables, used for power calculation
STA file	Static Timing Analysis file containing instance specific minimum/maximum transition times, timing windows and clock network data
SPEF file	Standard Parasitic Exchange File. Instance-specific signal wire parasitic data
Technology file (*.tech*)	Design-specific file containing metal layer information
Global System Requirement file (*.gsr*)	Design-specific information for operating conditions and analysis parameters
Pad files (*.ploc*, *.pad*,or *.pcell*)	Power and ground pad descriptions, cell names, instance names, and X,Y locations
APL	Apache Power Libraries. Contains electrical model for various components in the design


Inputs
Apache Power Library
Current Profiles
Instance Power Resistance
Intrinsic Decap
Outputs
Library and Technology-specific
Physical Library
LEF / LIB files
Technology
Apache Tech file
RedHawk
Power Calculation
Power Grid RLC Extraction
Dynamic Voltage Hotspot Analysis and Fixing
Vectorless Dynamic Sim
Impact on Timing
Dynamic Pwr / Volt Contour Maps
Dynamic Pwr / Volt Text Reports
Decap Maps
Warnings/ Violations
Design-specific
Physical Design
DEF / GDS files
Design & Analysis Spec
GSR file
Package Specs
.pcell / .ploc / .pad files
Package Parasitics
Parasitic Loading
SPEF / DSPF files
Instance Slew/Timing
STA file

4 Use Case(UC1): Grid Weakness Analysis

Using RedHawk, the user can verify the basic grid quality. In this analysis, the user can find:

1. Unconnected instances from power and ground rail
2. Unconnected from both power and ground rails
3. Weakly connected instances, i.e., instances that have high resistance from the PG bumps

This analysis can be done with minimal set of data. In this analysis, the tool requires only layout data and technology collaterals and power ground rail names and pad locations. RedHawk™ performs the power and ground network extraction. It identifies the relatively weak instances based on Shortest Path Resistance from power and ground pads and provides a Grid weakness map and report for users to review and debug design weakness. RedHawk™ also identifies instances that are not physically connected to the power/ground network in this process.

To do this analysis you need to specify all the required inputs to the tool to setup the design and specify the analysis condition in a configuration file and execute the flow. RedHawk™ provides violations as color coded maps overlaid on layout or as a report with the co-ordinates of violations for users to fix the design weaknesses. Here is the flow diagram describing the work-flow for Grid Weakness Analysis

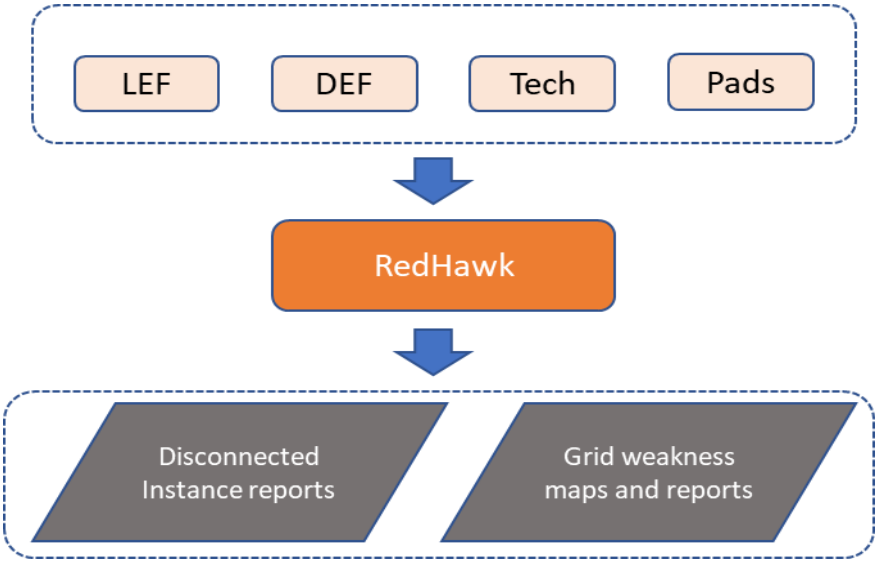


Figure 1 Work flow for Grid Weakness Analysis

5 Use Case(UC2): Static IR drop Analysis

Using RedHawk™, the user can perform static IR drop analysis. In this analysis, the user can find:

1. High Voltage drops on Instances
2. High voltage drops on power/ground interconnects

In this analysis, the tool requires only layout data and technology collaterals and power ground rail names and pad(voltage source) locations and Power constraints. The power constraints can be provided by a variety of ways. The scope of this document restricted to the work-flow and does not cover the exact specification. Users are advised to refer to RedHawk™ User Manual and consult with ANSYS Tool Applications Engineer for further advice on the different methods to specify.

RedHawk™ performs the power and ground network extraction. It assigns the current on the instances based on the power constraints provided by the user and performs a DC simulation also known as Static IR drop analysis to determine Instances and interconnects having excessive voltage drop issues. User can provide IR drop tolerance thresholds for violation reporting. To do this analysis you need to specify all the required inputs to the tool to setup the design and specify the analysis condition in a configuration file and execute the flow. RedHawk™ provides violations as color coded maps overlaid on layout or as a report with the co-ordinates of violations for users to fix the design weaknesses.

Here is the flow diagram describing the work-flow for static IR drop analysis.

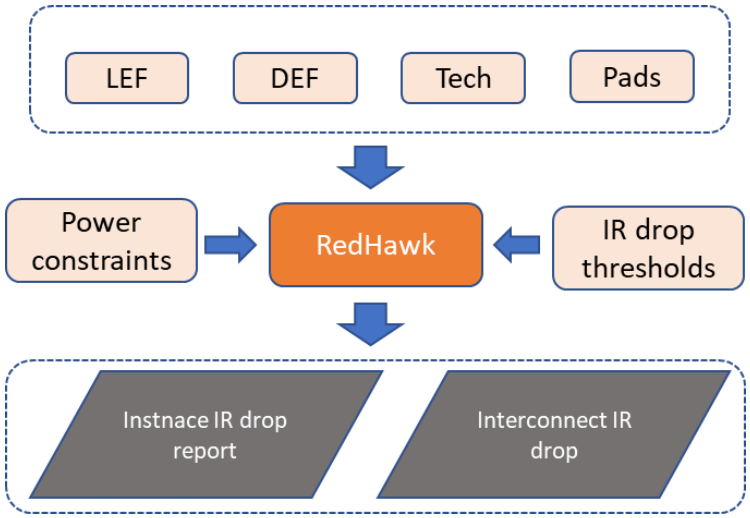


Figure 2 Work flow for Static IR drop Analysis

6 Use Case(UC3): Dynamic voltage drop Analysis

Using RedHawk™, the user can perform dynamic voltage drop analysis. In this analysis, the user can find:

1. High transient Voltage drops on Instance
2. High transient voltage drops on power/ground interconnects

In this analysis, the tool requires only layout data and technology collaterals and power ground rail names and pad(voltage source) locations and Power constraints and APL libraries to model the current demand when a cell switches or a capacitive when it does not switch.

RedHawk™ performs the power and ground network extraction. RedHawk™ determines the switching scenario (switching instances – which/when/how and non-switching instances) and models the instances based on the APL library. RedHawk™ then runs transient simulation also known as dynamic voltage drop analysis to determine Instances and interconnects having excessive voltage drop issues. User can provide IR drop tolerance thresholds for violation reporting. To do this analysis you need to specify all the required inputs to the tool to setup the design and specify the analysis condition in a configuration file and execute the flow. RedHawk™ provides violations as color coded maps overlaid on layout or as a report with the co-ordinates of violations for users to fix the design weaknesses.

Here is the flow diagram describing the work-flow for dynamic voltage drop analysis.

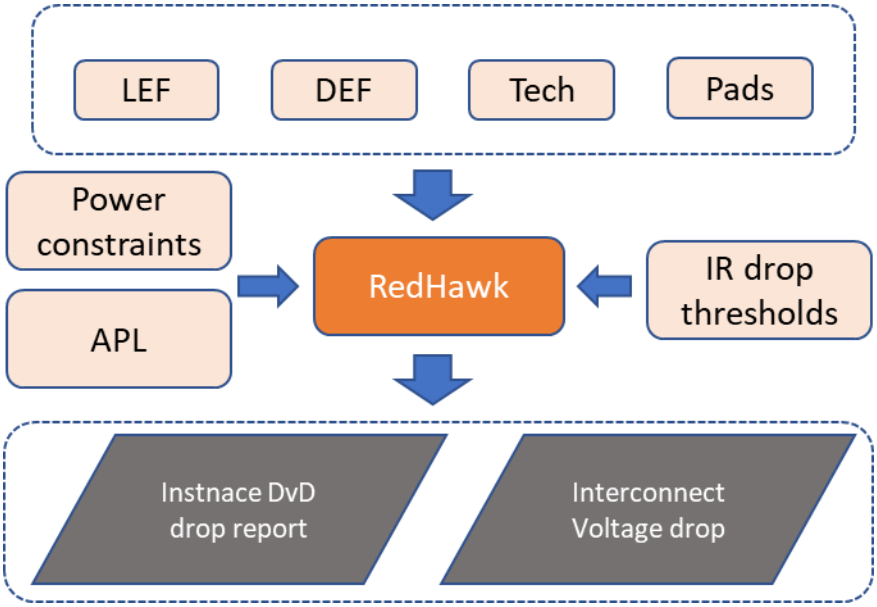


Figure 3 Dynamic voltage drop analysis

7 Use Case(UC4): Power Electromigration Analysis

Using RedHawk™, the user can perform Electromigration analysis on power and ground interconnects. In this analysis, the user can find high EM violations on the power and ground interconnects. The Electromigration analysis can be performed with DC or transient current profiles. If global power is assigned to the block similar to “Static IR analysis” mentioned in section 5, only average EM violations can be verified by the tool. If simulation current profiles are provided all 3 EM modes(average, rms and peak) can be verified by RedHawk™. The workflow for EM analysis can be combined with static IR or dynamic voltage drop analysis flow. It is typically recommended to perform EM analysis at high power, worst case corner and high temperature. The workflow for EM analysis is described below.

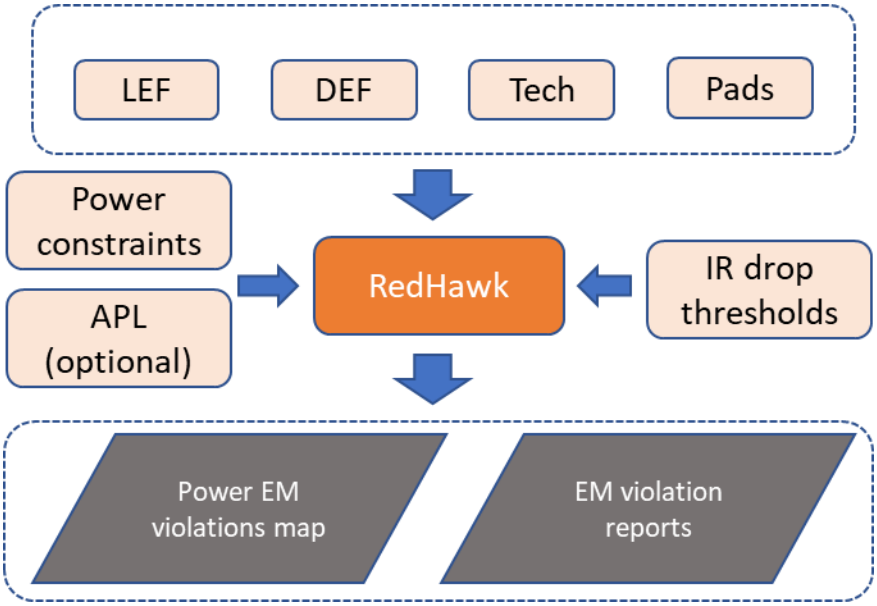


Figure 4 Power EM analysis work flow

8 Use Case(UC5): Signal Electromigration Analysis

Using RedHawk™, the user can perform Electromigration analysis on signal nets. In this analysis, the user can find high EM violations on the signal interconnects. This analysis, all 3 EM modes can be verified by RedHawk™. The work flow for EM analysis is described below.

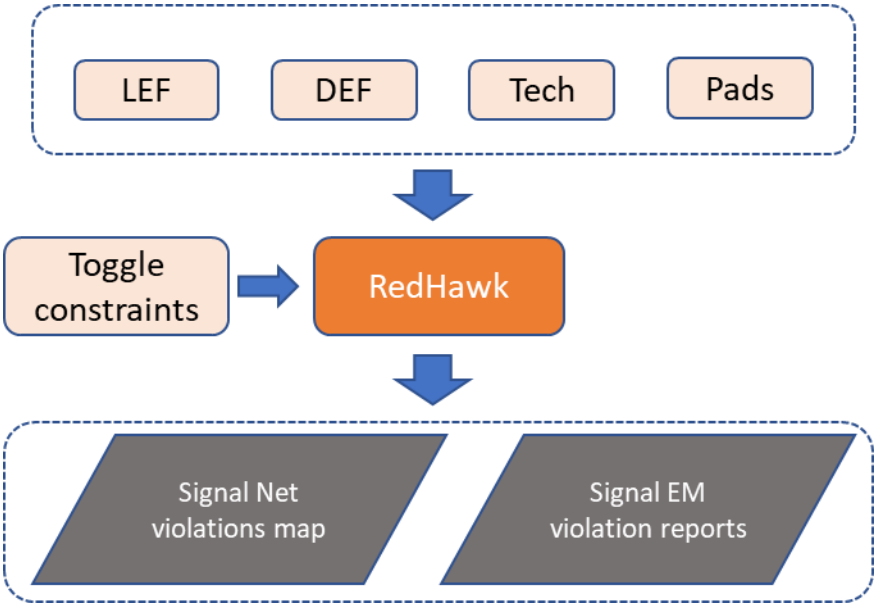


Figure 5 Signal EM analysis work flow

9 Error Scenario

9.1 Failure Mode and Effect Analysis

Use Case	Error Condition	Potential Malfunctions and Erroneous reporting	Tool Impact (TI)	**Justification for TI**	Tool Error Detection (TD)	Justification for TD	Error Detection and Prevention Methods	Tool Classification Level (TCL)
UC1	EC1	Tool does not identify Instances that are weakly connected to the power and ground rail	TI2	Missed design weakness	TD1	Error can be detected by additional checks	Please refer to section 10.1.1 for details	TCL1
UC1	EC2	Tool does not identify any disconnected Instances or interconnects	TI2	Missed design weakness	TD1	Error can be detected by additional checks	Please refer to section 10.1.2 for details	TCL1
UC2	EC3	Unreasonably large IR drop violations due to inputs, constraints or tool issues.	TI2	Wrong analysis results in tool reports can lead to undetected IR-Drop issues	TD1	Error can be detected by additional checks	Please refer to section 10.2.1 for details	TCL1
UC2	EC4	Very low or no IR violations reported by the tool	TI2	Wrong analysis results in tool reports can lead to undetected IR-Drop issues	TD1	Error can be detected by additional checks	Please refer to section 10.2.2 for details	TCL1

UC3	EC5	Unreasonably large IR drop violations due to inputs, constraints or tool issues.	TI2	Wrong analysis results in tool reports can lead to undetected IR-Drop issues	TD1	Error can be detected by additional checks	Please refer to section 10.3.1 for details	TCL1
UC3	EC6	Very low or no IR violations reported by the tool	TI2	Wrong analysis results in tool reports can lead to undetected IR-Drop issues	TD1	Error can be detected by additional checks	Please refer to section 10.3.2 for details	TCL1
UC4	EC7	Unreasonably high EM violations or tool was unable to identify any violations possibly due to constraints	TI2	Wrong analysis results in tool reports can lead to undetected EM weakness on power grid	TD1	Error can be detected by additional checks	Please refer to section 10.4.1 for details	TCL1
UC5	EC8	No violations or high EM violations	TI2	Wrong analysis results in tool reports can lead to undetected EM weakness on signal nets	TD1	Error can be detected by additional checks	Please refer to section 10.5.1 for details	TCL1

Table 1 Failure Mode and Effect Analysis

10 Error Detection and Prevention Methods

10.1 Use Case (UC1)– Grid weakness analysis

10.1.1 Error Condition (EC1)

To avoid design failures due to RedHawk™ not identifying and not reporting instances connected weakly to the power grid, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis.
2. Add margin to the power grid design
3. Perform static IR or DVD analysis to detect design weakness. Design weakness may manifest as higher IR drop violations in the Static IR or DVD analysis.
4. Perform silicon validation, to ensure the target frequency is met.

10.1.2 Error Condition (EC2)

To avoid design failures due to RedHawk™ missing to identify or report any unconnected instances in the design, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis.
2. Perform LVS checks to ensure that design intent in schematic is captured in layout and there are no disconnects reported in the LVS check

10.2 Use Case (UC2) – Static IR drop analysis

10.2.1 Error Condition (EC3)

To avoid design failures due to RedHawk™ missing to identify or report any instances of High Static IR drop, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis.
2. Add margin to the power grid design
3. Perform DVD analysis to check if there are any design weakness. Design weakness may manifest as higher IR drop violations in the DVD analysis.

10.2.2 Error Condition (EC4)

To avoid design failures due to RedHawk™ finding and reporting a very low or no IR drop, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis.
2. Perform dynamic voltage drop (DVD) analysis to detect design weakness. Design weakness may manifest as higher IR drop violations in the DVD analysis.
3. Perform silicon validation, to ensure the target frequency is met

10.3 Use Case (UC3) – Dynamic voltage drop analysis

10.3.1 Error Condition (EC5)

To avoid design failures due to RedHawk™ finding and reporting unreasonably high IR drop violations, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis
2. Add margin to the power grid design
3. Perform silicon validation, to ensure the target frequency is met

10.3.2 Error Condition (EC6)

To avoid design failures due to RedHawk™ finding and reporting unreasonably low or no IR drop violations, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis.
2. Add margin to the power grid design
3. Perform silicon validation, to ensure the target frequency is met and the functionality of the silicon is as expected

10.4 Use Case (UC4) – Power grid Electromigration analysis

10.4.1 Error Condition (EC7)

To avoid design failures due to RedHawk™ finding and reporting unreasonably high EM violations or not generating any EM violations on Power/ground nets, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis.
2. Build more redundancy in power grid
3. Perform silicon validation burn-in testing, to ensure the target frequency is met and the functionality of the silicon is as expected

10.5 Use Case (UC4) – Signal Electromigration analysis

10.5.1 Error Condition (EC8)

To avoid design failures due to RedHawk™ finding and reporting unreasonably high EM violations or not generating any EM violations on signal nets, it is recommended to perform the following detection and prevention measures:

1. Verify if the tool inputs and constraints provided are appropriate for the design and fix them if necessary and redo the analysis
2. Perform validation with spice simulations to ensure that there are no EM violations.
3. Perform silicon validation burn-in testing, to ensure the functionality of the silicon is as expected.